INFOSYS MOCK TEST 2

Moderate to Hard

Rules and Instructions

- Attempt every section within its suggested time.
- Do not use calculators, notes, search engines or AI.
- Select one best answer for each MCQ.
- No negative marking is assumed for this practice test.
- Explain your approach and complexity during coding and interview rounds.
- These are original practice questions based on common fresher recruitment topics, not an official Infosys paper.

ROUND 1: ONLINE ASSESSMENT

Section A: Logical Reasoning

10 Questions | 15 Minutes

1. Find the next term: 4, 9, 19, 39, 79, ...
   A. 119 B. 139
   C. 159 D. 169

2. If TABLE is coded as UBCMF, how is CHAIR coded?
   A. DIBJS B. DIBIR
   C. DHBJS D. DIBJT

3. All analysts are graduates. Some graduates are programmers. Which conclusion definitely follows?
   A. All analysts are programmers B. Some programmers are analysts
   C. All analysts are graduates D. No programmer is a graduate

4. A is the brother of B. B is the mother of C. How is A related to C?
   A. Father B. Maternal uncle
   C. Brother D. Grandfather

5. A person faces west, turns 90 degrees clockwise, then 180 degrees anticlockwise. Which direction now?
   A. North B. South
   C. East D. West

6. In a group of 60, 35 know Java, 30 know Python, and 15 know both. How many know neither?
   A. 5 B. 10
   C. 15 D. 20

7. Find the missing pair: BC, EF, HI, KL, ...
   A. MN B. NO
   C. OP D. QR

8. Six people sit around a circular table. A is opposite D; B is immediately clockwise of A. Who is immediately anticlockwise of A?
   A. C B. E
   C. F D. Cannot be determined

9. If 1 January is Monday in a non-leap year, what day is 1 March?
   A. Wednesday B. Thursday
   C. Friday D. Saturday

10. A server was unavailable; therefore, a user could not log in. Which assumption is most relevant?
   A. Login requires the server B. Users never forget passwords
   C. The network is always fast D. The user has no account

Section B: Mathematical Ability

10 Questions | 25 Minutes

1. A price is increased by 25% and then reduced by 20%. Net change is:
   A. No change B. 5% increase
   C. 5% decrease D. 10% decrease

2. A and B together finish a job in 8 days. A alone takes 12 days. B alone takes:
   A. 18 days B. 20 days
   C. 24 days D. 30 days

3. A 180 m train crosses a pole in 9 seconds. Speed is:
   A. 60 km/h B. 64 km/h
   C. 72 km/h D. 80 km/h

4. Compound interest on Rs. 10,000 at 10% per annum for 2 years, compounded annually, is:
   A. Rs. 2,000 B. Rs. 2,050
   C. Rs. 2,100 D. Rs. 2,200

5. If x+y=14 and xy=45, find $x^2 + y^2$.
   A. 96 B. 100
   C. 106 D. 116

6. A committee of 2 is selected from 6 people. Number of ways:
   A. 12 B. 15
   C. 18 D. 30

7. A bag has 5 white, 4 black and 3 red balls. Probability of not selecting a black ball is:
   A. 1/3 B. 1/2
   C. 2/3 D. 3/4

8. A boat travels 24 km downstream in 2 hours and the same distance upstream in 3 hours. Speed of the stream is:
   A. 1 km/h B. 2 km/h
   C. 3 km/h D. 4 km/h

9. The median of 3, 7, 8, 11, 14, 19, 21 is:
   A. 8 B. 11
   C. 14 D. 19

10. A 40-litre solution contains 25% acid. How much water must be added to make it 20% acid?
   A. 5 L B. 8 L
   C. 10 L D. 12 L

Section C: Verbal Ability

10 Questions | 12 Minutes

1. Choose the synonym of Meticulous.
   A. Careless B. Thorough
   C. Hasty D. Ordinary

2. Choose the antonym of Obsolete.
   A. Outdated B. Ancient
   C. Current D. Unused

3. Choose the grammatically correct sentence.
   A. Neither of the answers are correct. B. Neither of the answers is correct.
   C. Neither answers is correct. D. Neither answer are correct.

4. Had I known about the delay, I _____ earlier.
   A. will leave B. would have left
   C. left D. would leave

5. The new policy will _____ employee productivity.
   A. effect B. affect
   C. effects D. affects

6. Choose the correctly punctuated sentence.
   A. However I completed the task. B. However, I completed the task.
   C. However I, completed the task. D. However; I, completed the task.

7. Choose the passive form: The manager will review the report.
   A. The report will be reviewed by the manager. B. The report is reviewed by the manager.
   C. The manager will be reviewed by the report. D. The report has reviewed the manager.

8. The solution was expensive; _____, it was reliable.
   A. therefore B. nevertheless
   C. because D. unless

9. Choose the word closest to Ambiguous.
   A. Clear B. Uncertain
   C. Exact D. Obvious

10. Identify the error: The quality of these products have improved.
    A. quality B. these
    C. have D. improved

Section D: Pseudocode and Output Tracing

5 Questions | 10 Minutes

1. Output?
   `s=0; for i=1 to 5:  if i mod 2==1:  s=s+i; print(s)`
   A. 6 B. 8
   C. 9 D. 15

2. Output?
   `f(n):  if n<=1 return 1; return n*f(n-1); print(f(4))`
   A. 10 B. 16
   C. 24 D. 120

3. Output?
   `a=[1,2,3,4]; for i=0 to 2:  a[i+1]=a[i+1]+a[i]; print(a[3])`
   A. 7 B. 8
   C. 9 D. 10

4. What is the time complexity of two nested loops, each running from 1 to n?
   A. $O(n)$ B. $O(\log n)$
   C. $O(n^2)$ D. $O(2^n)$

5. Output?
   `x=10; y=3; while x>y:  x=x-y; print(x)`
   A. 0 B. 1
   C. 2 D. 3

Section E: Puzzle Solving

4 Questions | 8 Minutes

1. A, B, C and D each have a different role: tester, developer, analyst and designer. A is not developer or tester. B is not analyst. C is designer. D is not tester. If A is analyst, determine the remaining roles.

2. You have 3 switches outside a closed room and one bulb inside. You may operate switches as you wish but enter the room only once. How do you identify the correct switch?

3. A two-digit number has digit sum 11. Reversing its digits increases it by 27. Find the number.

4. Five tasks must be scheduled. A must occur before C; B must occur after D; E must occur before A. Give one valid order.

Section F: English Writing

10 Minutes
Write 180–220 words on:
Should companies prioritise technical skills or communication skills when hiring fresh graduates?
Present both sides and give a balanced conclusion.

ROUND 2: CODING ASSESSMENT

Suggested Time: 60 Minutes

1. **Longest Substring Without Repeating Characters**

   Given a string, return the length of the longest substring containing no repeated character.

   Example:

   Input: `abcabcbb`

   Output: `3`

   Target: Sliding window approach with O(n) time complexity.

2. **Subarray Sum Equals K**

   Given an integer array, which may contain negative values, and an integer K, count contiguous subarrays whose sum equals K.

   Example:

   Array: `[1,1,1]`

   K = 2

   Output: `2`

   Target: Prefix sums with a hash map, expected O(n).

ROUND 3: TECHNICAL INTERVIEW

1. Explain the difference between JDK, JRE and JVM.
2. What is the difference between ArrayList and LinkedList?
3. Explain HashMap internally at a high level.
4. What happens when two keys hash to the same bucket?
5. Explain checked versus unchecked exceptions.
6. What is garbage collection in Java?
7. Explain recursion and when iteration may be preferable.
8. Compare BFS and DFS. Give one use case for each.
9. Explain merge sort and quicksort.
10. Compare their average and worst-case complexities.
11. What is a binary search tree?
12. Explain inorder traversal.
13. What is a database index? What trade-off does it introduce?
14. Write SQL to find the second-highest salary from an Employee table.
15. Explain INNER JOIN versus LEFT JOIN.
16. What is a database transaction? Explain commit and rollback.
17. What is context switching?
18. Explain deadlock prevention versus deadlock avoidance.
19. What is the difference between TCP and UDP?
20. Explain DNS resolution at a high level.
21. What is REST? Explain GET, POST, PUT, PATCH and DELETE.
22. How would you secure a web API?
23. Explain authentication, authorization, validation and rate limiting.
24. Explain your strongest project architecture, database design, APIs, testing and deployment.

ROUND 4: HR AND MANAGERIAL INTERVIEW

1. Why Infosys rather than another IT services company?
2. What do you understand by client-facing work?
3. Describe a time you took ownership without being asked.
4. Tell me about a time you received difficult feedback.
5. How would you handle a teammate who is not contributing?
6. If assigned a technology you do not know, how would you deliver the task?

7. How do you prioritise quality when the deadline is close?
8. Are you willing to work in shifts if the project requires it?
9. Are you comfortable with relocation and training requirements stated in the offer?
10. What motivates you to learn continuously?
11. What would you do if you made a production mistake?
12. Where do you see yourself in three to five years?

ANSWER KEY

Logical Reasoning:
1-C, 2-A, 3-C, 4-B, 5-B, 6-B, 7-B, 8-D, 9-B, 10-A.
Mathematical Ability:
1-A, 2-C, 3-C, 4-C, 5-C, 6-B, 7-C, 8-B, 9-B, 10-C.
Verbal Ability:
1-B, 2-C, 3-B, 4-B, 5-B, 6-B, 7-A, 8-B, 9-B, 10-C.
Pseudocode:
1-C, 2-C, 3-D, 4-C, 5-A.
Puzzle Guidance:

1. B=tester and D=developer.
2. Turn switch 1 on for several minutes, turn it off, then turn switch 2 on. Enter the room: lit bulb = switch 2; unlit but warm bulb = switch 1; unlit and cold bulb = switch 3.
3. The number is 47.
4. One valid order is D, B, E, A, C.